

L.D. COLLEGE OF ENGINEERING, AHMEDABAD

Department of Information Technology

Data Science (BE05016021)

Project Based Learning (PBL) Activity Report

Intelligent Customer Segmentation and Marketing Campaign Optimization System

*Customer segmentation, campaign response prediction and an interactive decision-support
dashboard using the Customer Personality Analysis dataset*

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Abstract

Retail companies often run marketing campaigns without a clear understanding of which customers are likely to respond, which leads to inefficient spending and poor targeting. This project develops a data-driven customer intelligence system using the Customer Personality Analysis dataset (2,240 customers, 29 attributes). After cleaning and feature engineering (2,054 customers retained), descriptive analytics is used to understand customer demographics, spending and purchase channels. Sampling and estimation techniques provide confidence intervals for key business measures and statistically test the differences between responders and non-responders. A multiple linear regression model predicts customer spending ($R^2 = 0.71$). K-Means clustering identifies four customer segments – Premium Spenders, Established Families, Budget Families and Young Starters – showing that half of the customers generate about 90% of revenue. Logistic Regression, Decision Tree and Random Forest classifiers are trained to predict campaign response; Logistic Regression performs best with a test ROC-AUC of 0.885 and a recall of 81%. A targeting simulation shows that contacting only the top 30% of customers ranked by the model captures about 80% of responders with 70% fewer contacts, turning a loss-making campaign into a profitable one. All results are delivered through an interactive Streamlit dashboard that allows business users to explore segments, evaluate models, score new customers and download campaign target lists.

Keywords: customer segmentation, K-Means, logistic regression, random forest, campaign response prediction, confidence interval, stratified sampling, Streamlit dashboard

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1. Introduction

1.1 Background – A Real-World Case Study

Marketing is one of the largest costs for a retail business. In a traditional mass campaign every customer receives the same offer through mail, phone, catalog or e-mail, and the company pays for every contact. In the company studied in this project, only about 15% of customers accepted the most recent campaign, which means roughly 85% of the contact cost produced no sales. At the same time, customers are very different from each other – in income, family situation, spending habits and preferred channels – so a single offer cannot suit everyone.

Data science allows a company to move from mass marketing to **targeted, personalised marketing**: understanding who its customers are (segmentation), predicting who will respond (classification) and estimating how valuable each customer is (regression). This project applies these techniques to a real retail dataset and packages the results as a decision-support application for marketing teams.

1.2 Problem Statement

To analyse customer demographic, behavioural and purchase data in order to identify meaningful customer segments and to build a classification model that predicts customer response to marketing campaigns, so that the company can target the right customers with the right offers and reduce wasted marketing spend. An interactive dashboard must present segments, model predictions and campaign insights to business stakeholders.

1.3 Objectives

1. Clean and prepare the customer data, and engineer meaningful features.
2. Perform descriptive analytics (EDA) to understand customer behaviour and campaign performance.
3. Apply sampling and estimation: confidence intervals, random vs stratified sampling, bootstrap, sample-size planning and hypothesis testing.
4. Build a linear regression model to estimate customer spending.
5. Segment customers using K-Means clustering and recommend a marketing strategy for each segment.
6. Build and compare Logistic Regression, Decision Tree and Random Forest models to predict campaign response.
7. Quantify the business impact of model-based targeting.
8. Develop an interactive Streamlit dashboard for business stakeholders.

1.4 Scope of the Project

The project covers the complete data science life cycle – data understanding, preparation, analysis, modelling, evaluation and deployment as an application. It uses one historical dataset from a single retail company; real-time data integration with a CRM system is outside the scope and is discussed as future work.

2. Literature Review

Customer segmentation and response modelling are well-studied problems in marketing analytics. The following works form the background for the methods used in this project.

- **Market segmentation (Wedel & Kamakura, 2000):** Describe segmentation as dividing a heterogeneous market into homogeneous groups that respond similarly to marketing actions, and review clustering-based methods. This supports the use of behavioural and demographic variables to form actionable segments.
- **RFM analysis (Hughes, 1994):** Introduced Recency, Frequency and Monetary value as the key indicators of customer value in database marketing. Recency, purchase counts and total spending engineered in this project follow the same idea, and Recency indeed emerged as one of the strongest predictors of response.
- **Data mining in CRM (Ngai, Xiu & Chau, 2009):** Reviewed data-mining applications in customer relationship management and found clustering and classification to be the most frequently used techniques for customer identification, attraction and retention – the two techniques combined in this project.
- **Campaign response prediction (Moro, Cortez & Rita, 2014):** Predicted the success of bank telemarketing calls with data-driven models and showed that ranking customers by predicted probability and contacting only the top fraction greatly increases campaign efficiency – the approach used in the targeting simulation of this project.
- **K-Means and cluster validation (MacQueen, 1967; Rousseeuw, 1987):** K-Means partitions observations into k clusters by minimising within-cluster variance; the silhouette coefficient measures how well each observation fits its cluster. Both are used to build and validate the customer segments.
- **Random Forests (Breiman, 2001):** Proposed an ensemble of decorrelated decision trees that reduces the overfitting of single trees and provides feature importance – used here as the ensemble model compared with Logistic Regression and a single Decision Tree.
- **Learning from imbalanced data (He & Garcia, 2009):** Showed that accuracy is misleading on imbalanced data and recommended cost-sensitive learning and metrics such as recall, F1 and ROC-AUC. This project uses class weighting and these metrics because only 15% of customers respond.

Research gap addressed: many studies focus on either segmentation or response prediction. This project combines both – the segment produced by clustering is used as an input to the response model – and connects the model output directly to a business decision (how many customers to contact and the resulting profit) through an interactive application.

3. Dataset Description

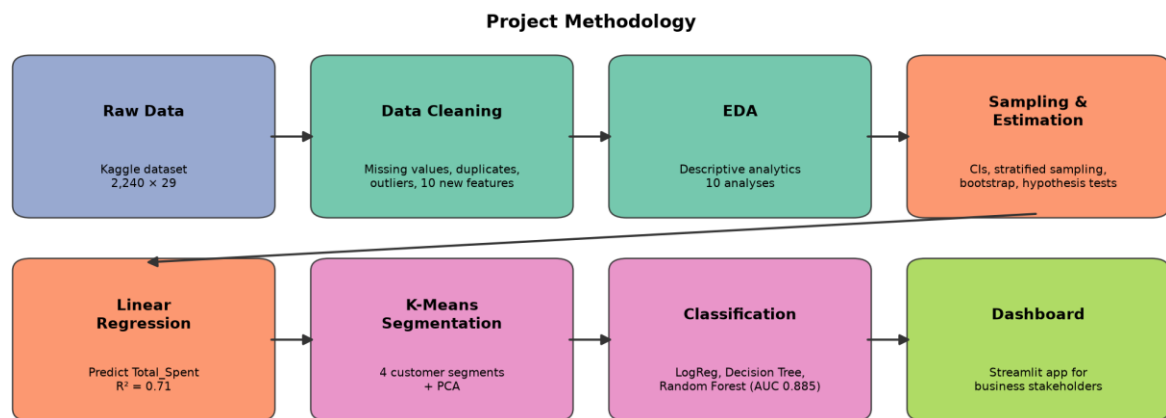
Table 1: Dataset overview

Property	Details
Name	Customer Personality Analysis
Source	Kaggle – kaggle.com/datasets/imakash3011/customer-personality-analysis
Size	2,240 customers × 29 attributes (tab-separated file)
Target variable	Response – 1 if the customer accepted the offer in the last campaign (15% positive)

Table 2: Dataset attributes

Group	Columns	Description
Customer profile	ID, Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome, Dt_Customer, Recency, Complain	Demographics, household, enrolment date, days since last purchase, complaints
Spending (2 years)	MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds	Amount spent per product category
Purchase channels	NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth	Purchases by channel and website visits
Campaign response	AcceptedCmp1–AcceptedCmp5, Response	Acceptance of the five earlier campaigns and the last campaign
Others	Z_CostContact, Z_Revenue	Constant columns (cost 3 per contact, revenue 11 per response)

4. Methodology

*Figure 1: Project methodology***Table 3: Mapping of project steps to course topics**

Step	Course topic	Techniques
Data cleaning	Data preprocessing	Median imputation, duplicate and outlier removal, category grouping, feature engineering
EDA	Descriptive analytics	Distributions, group comparisons, correlation analysis
Sampling & estimation	Sampling and estimation	t- and z-confidence intervals, SRS vs stratified sampling, bootstrap, sample-size formula, t-test, chi-square test
Spending prediction	Linear regression	Simple and multiple linear regression, OLS significance, residual diagnostics, cross-validation
Segmentation	Clustering	Standardisation, K-Means, elbow method, silhouette score, PCA
Response prediction	Logistic regression & classification	Logistic Regression, Decision Tree, Random Forest, GridSearchCV, ROC-AUC, confusion matrix, gains/lift
Application	Software development	Streamlit dashboard with Plotly charts

Tools: Python 3.13, pandas, NumPy, SciPy, statsmodels, scikit-learn, Matplotlib, Seaborn, Plotly, Streamlit, Jupyter Notebook and Git/GitHub.

5. Data Preprocessing and Feature Engineering

5.1 Data Quality Issues and Treatment

Table 4: Data cleaning summary

Issue found	Treatment
182 duplicate customers (identical in all columns except ID)	Removed
24 missing values in Income	Filled with the median (robust to outliers)
Z_CostContact and Z_Revenue are constant; ID is only an identifier	Dropped
Impossible birth years (1893, 1899, 1900)	Removed customers born before 1940
Extreme income value (666,666)	Removed incomes above 600,000
Invalid marital labels (Absurd, YOLO, Alone)	Grouped into Partner (Married/Together) and Single
Five education levels	Grouped into Undergraduate, Graduate, Postgraduate

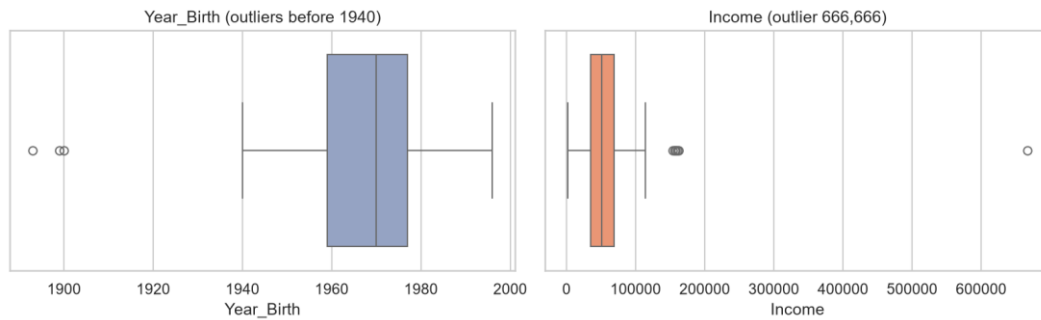


Figure 2: Outliers in Year_Birth and Income before cleaning

After cleaning, **2,054 customers** remain with no missing values. The cleaning logic is implemented once in `utils.py` so that the notebook and the dashboard use identical preprocessing.

5.2 Feature Engineering

Table 5: Engineered features

New feature	Definition
Age	2014 – Year_Birth (data collected in 2014)
Total_Spent	Sum of the six Mnt* spending columns
Children	Kidhome + Teenhome
Partner	1 if Married or Together
Family_Size	1 + Partner + Children
Is_Parent	1 if Children > 0
Total_Purchases	Deals + Web + Catalog + Store purchases
Total_Accepted_Cmp	AcceptedCmp1 + ... + AcceptedCmp5
Customer_Days	Days since enrolment (to the latest enrolment date)
Edu_Code	0 = Undergraduate, 1 = Graduate, 2 = Postgraduate

6. Exploratory Data Analysis

The average customer is 45 years old, earns about 52,000 per year and spent about 606 in two years. Spending is strongly right-skewed: most customers spend little while a small group spends a lot.



Figure 3: Distribution of Age, Income and Total Spending

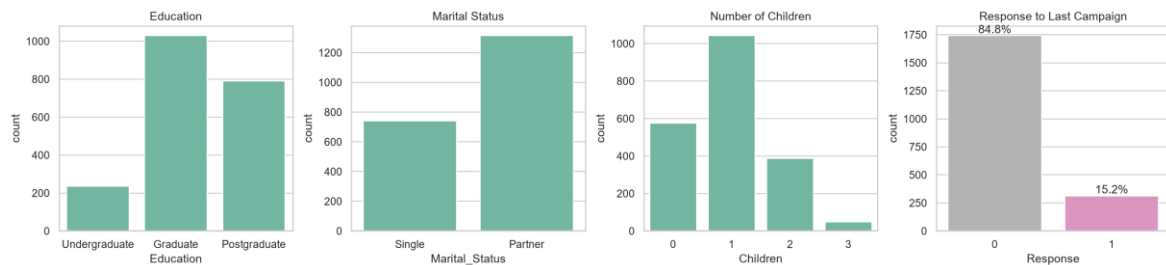


Figure 4: Customer profile and response to the last campaign

Only **15.2%** of customers responded to the last campaign, so the target classes are imbalanced (85:15).

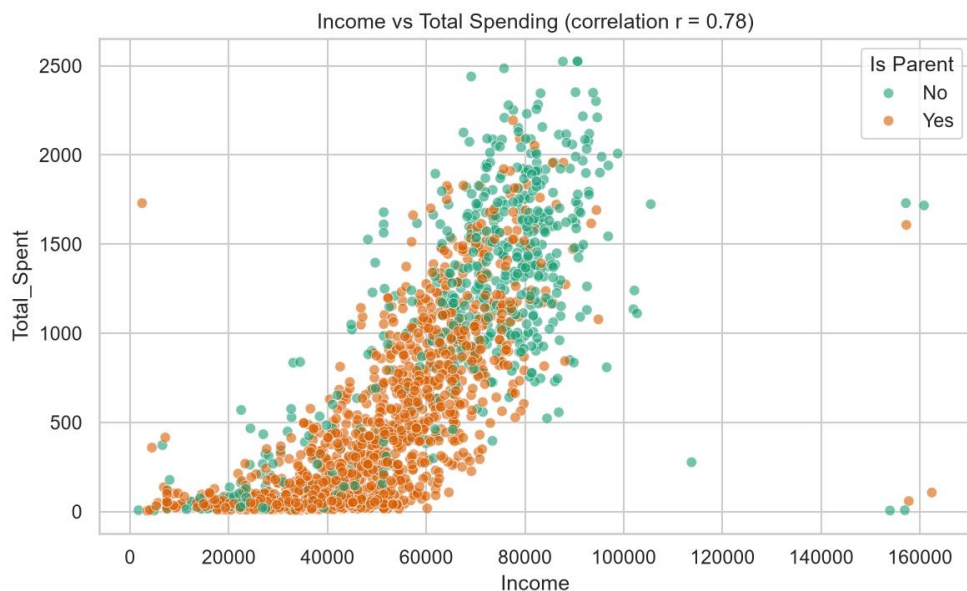


Figure 5: Income vs Total Spending ($r = 0.78$)

Income and spending have a strong positive relationship ($r = 0.78$). Customers without children dominate the high-spending region; non-parents spend about **6 times more** than parents (median 1,190 vs 202).

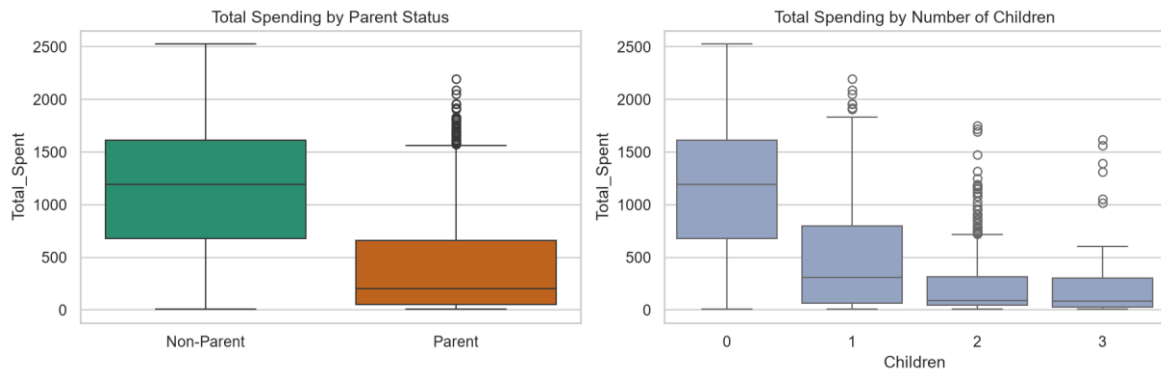


Figure 6: Spending of parents vs non-parents

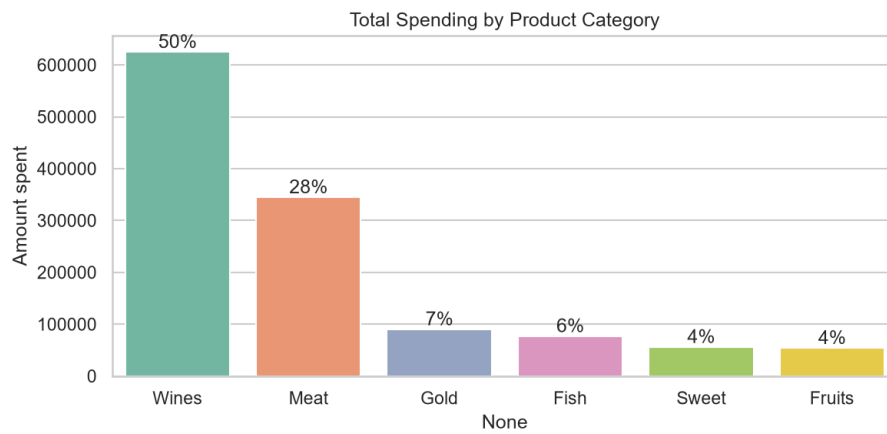


Figure 7: Spending by product category

Wine (50%) and meat (28%) account for about 78% of revenue. In-store is the main purchase channel (39% of purchases), followed by web (28%) and catalog (18%); 16% of purchases use a discount.

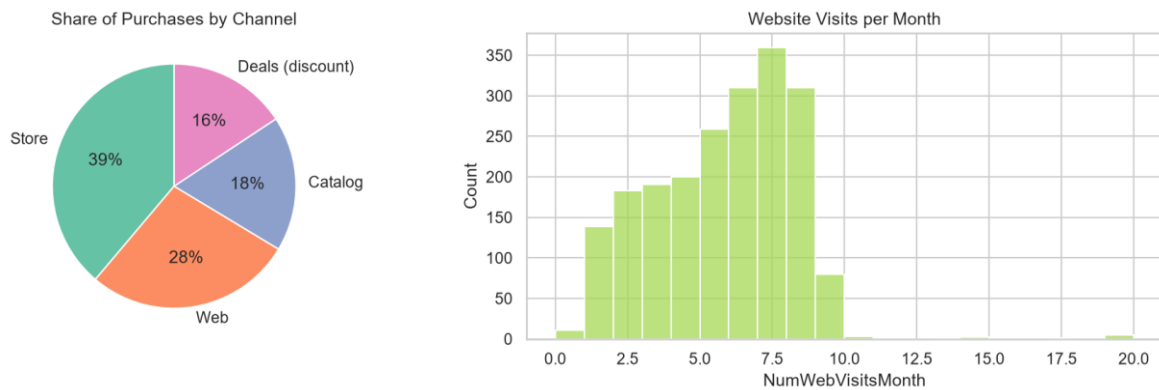


Figure 8: Purchase channels and website visits

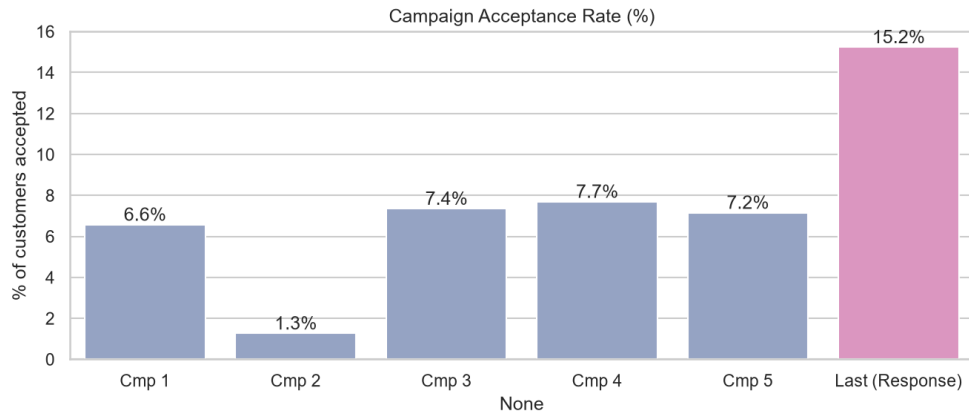


Figure 9: Acceptance rate of each campaign

Earlier campaigns had very low acceptance (1.3%–7.7%); campaign 2 almost completely failed. Even the most recent campaign was accepted by only 15.2% of customers.

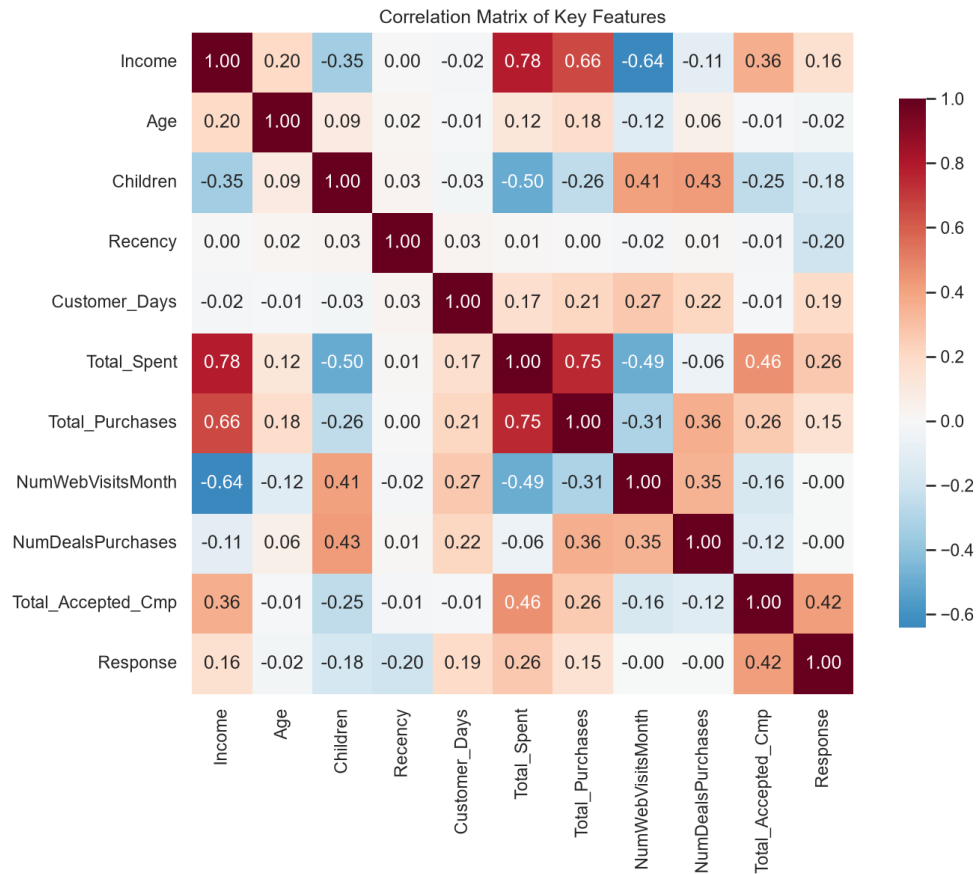


Figure 10: Correlation matrix of key features

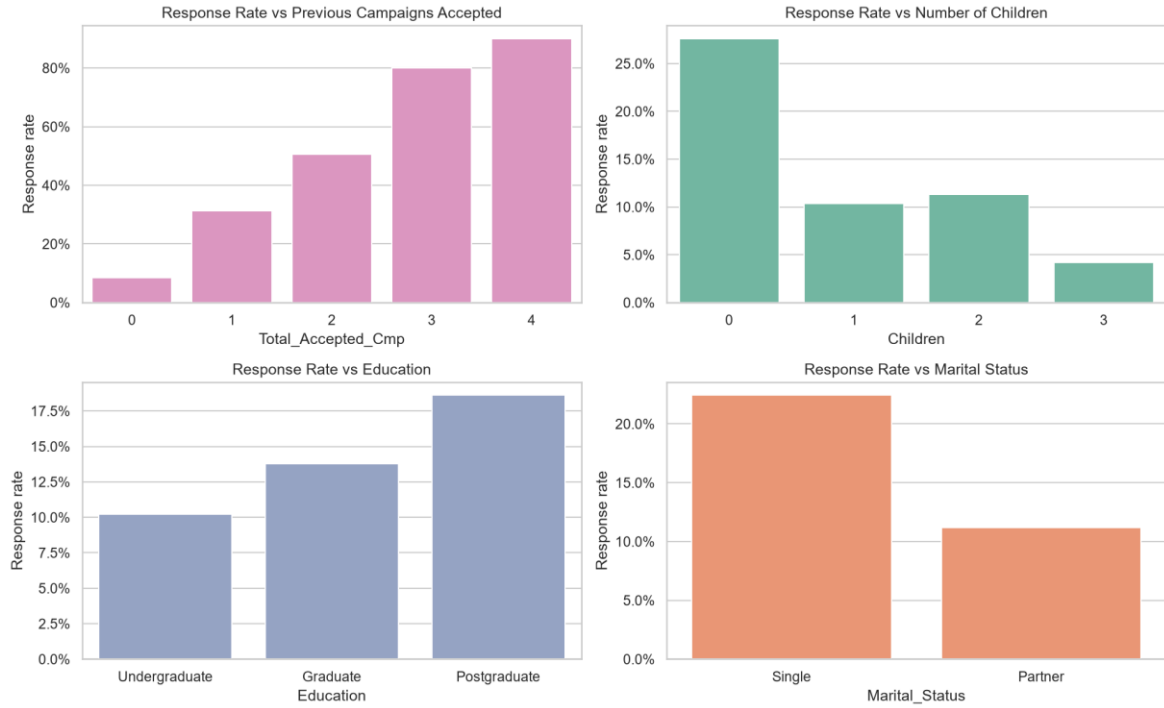


Figure 11: Response rate by previous acceptances, children, education and marital status

Response rises sharply with previous acceptances (8% for none, 31% for one, 51% for two and above 80% for three or more). Customers without children (28%), singles (22%) and postgraduates (19%) respond more often.

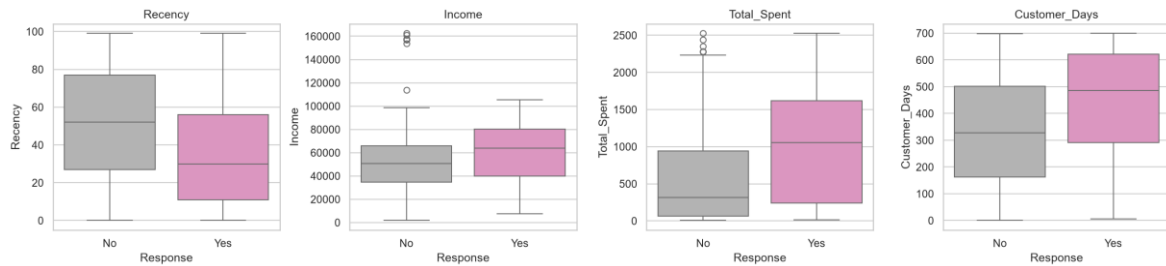


Figure 12: Responders vs non-responders

Responders bought more recently (35 vs 51 days), earn more ($\approx 60,000$ vs $\approx 51,000$), spend almost twice as much (≈ 980 vs ≈ 540) and have been customers for longer (≈ 443 vs ≈ 336 days).

7. Sampling and Estimation

7.1 Point Estimates and Confidence Intervals

For means, the t-interval $\bar{x} \pm t(\alpha/2, n-1) \cdot s/\sqrt{n}$ is used because the population standard deviation is unknown. For the response proportion, the z-interval $\hat{p} \pm z(\alpha/2) \cdot \sqrt{(\hat{p}(1-\hat{p})/n)}$ is used.

Table 6: Point estimates and 95% confidence intervals

Parameter	Point estimate	95% confidence interval
Mean Total_Spent	606.4	580.4 – 632.5
Mean Income	52,036	51,107 – 52,965

Mean Age	45.1	44.6 – 45.6
Mean Recency (days)	49.0	47.7 – 50.2
Response rate	15.2%	13.7% – 16.8%

7.2 Effect of Sample Size

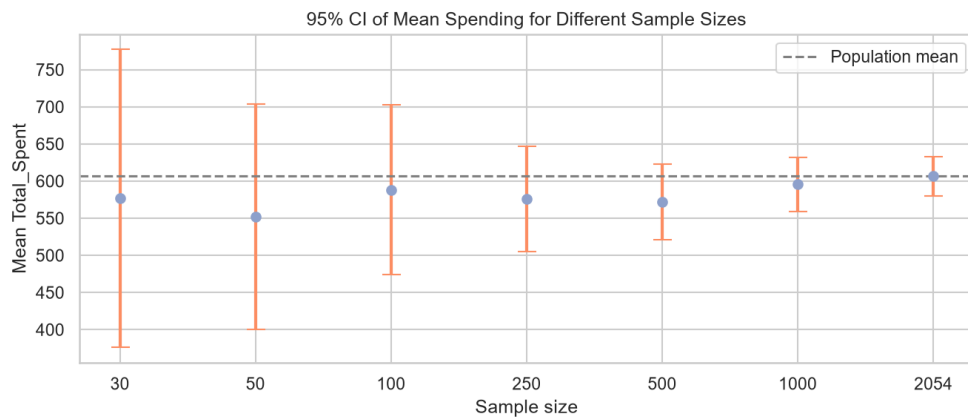


Figure 13: 95% confidence interval of mean spending for different sample sizes

The width of the confidence interval falls from 401 at $n = 30$ to 52 at $n = 2,054$, roughly in proportion to $1/\sqrt{n}$. Larger samples give more precise estimates.

7.3 Simple Random vs Stratified Sampling

Two hundred samples of 300 customers were drawn with each method. With simple random sampling, the response rate in a sample varied from 10% to 20% purely by chance (standard deviation 1.9 percentage points). Stratified sampling by Response kept it at 15.3% in every sample. For this reason the train/test split for the classification models is stratified.

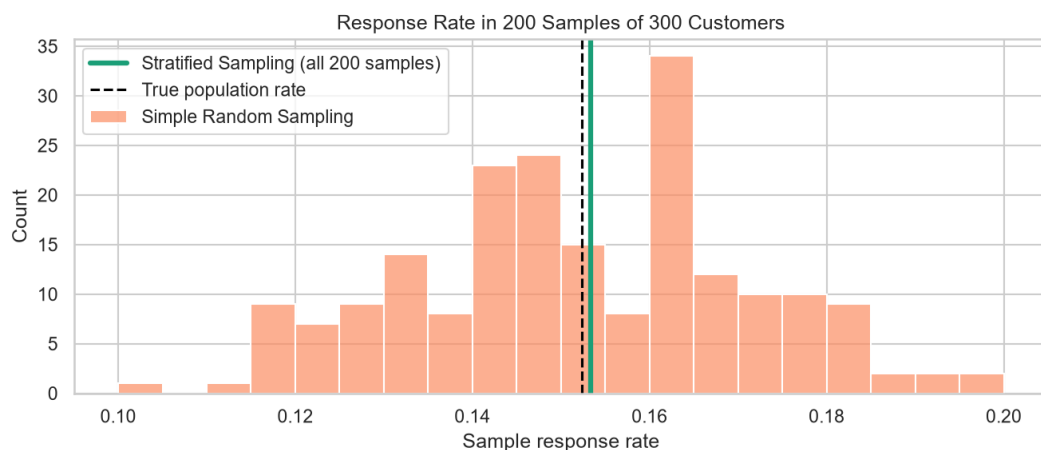


Figure 14: Response rate in 200 samples: simple random vs stratified sampling

7.4 Bootstrap and the Central Limit Theorem

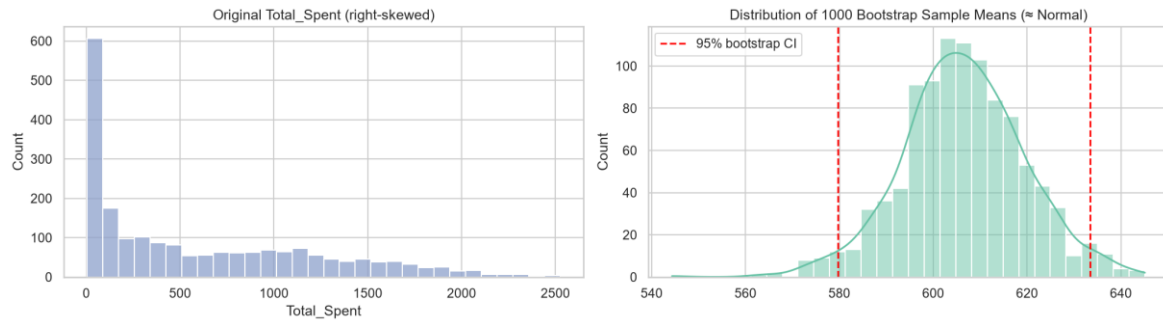


Figure 15: Original spending distribution and the distribution of 1,000 bootstrap means

Although spending is skewed, the distribution of bootstrap sample means is approximately normal, illustrating the Central Limit Theorem. The bootstrap 95% interval (579.7 – 633.5) closely matches the t-interval.

7.5 Sample Size Planning

Using $n = z^2 \cdot p(1-p)/E^2$ with $p = 0.152$, a pilot campaign needs about **552 customers** to estimate the response rate within $\pm 3\%$ at 95% confidence (199 for $\pm 5\%$, 1,241 for $\pm 2\%$).

7.6 Hypothesis Testing ($\alpha = 0.05$)

Table 7: Hypothesis test results

Variable	Test	Result	p-value	Decision
Total_Spent	Welch t-test	Responders 980 vs 539	< 0.001	Significant
Income	Welch t-test	60,127 vs 50,581	< 0.001	Significant
Recency	Welch t-test	35 vs 51 days	< 0.001	Significant
Customer_Days	Welch t-test	443 vs 336 days	< 0.001	Significant
Age	Welch t-test	44.5 vs 45.2 years	0.319	Not significant
NumWebVisitsMonth	Welch t-test	5.30 vs 5.33	0.887	Not significant
Education	Chi-square	$\chi^2 = 13.19$, df = 2	0.001	Associated
Marital_Status	Chi-square	$\chi^2 = 45.48$, df = 1	< 0.001	Associated
Is_Parent	Chi-square	$\chi^2 = 93.44$, df = 1	< 0.001	Associated
Complain	Chi-square	$\chi^2 = 0.00$, df = 1	1.000	Not associated

The tests confirm that responders differ significantly from non-responders in spending, income, recency and tenure, and that education, marital status and parenthood are associated with response. Age, website visits and complaints show no significant relationship.

8. Linear Regression – Predicting Customer Spending

The target is Total_Spent. The features are Income, Age, Children, Partner, Education, Recency, Customer_Days and NumWebVisitsMonth. The spending and purchase-count columns are excluded because Total_Spent is computed from them (data leakage). Data is split 80/20 into training and test sets.

Table 8: Linear regression performance

Model	Test R ²	RMSE	MAE	5-fold CV R ²
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Simple LR (Income only)	0.631	382.7	285.1	0.609
Multiple LR (8 features)	0.712	338.2	247.5	0.699

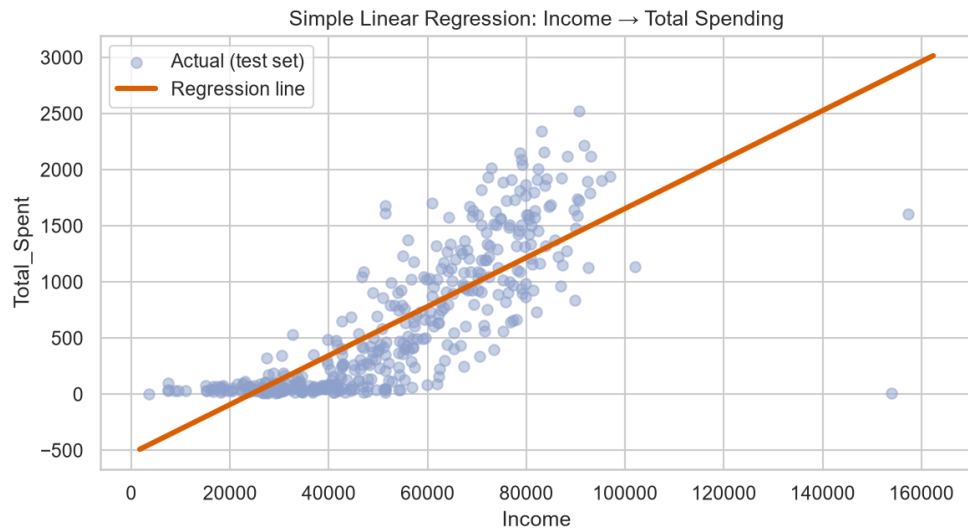


Figure 16: Simple linear regression: Income → Total Spending

Table 9: OLS coefficients (training set)

Feature	Coefficient	p-value	Significant
Income	0.020	< 0.001	Yes
Children	-216.6	< 0.001	Yes
Customer_Days	0.483	< 0.001	Yes
NumWebVisitsMonth	9.96	0.035	Yes
Recency	0.402	0.150	No
Partner	-8.14	0.629	No
Age	0.096	0.896	No
Edu_Code	-1.62	0.899	No

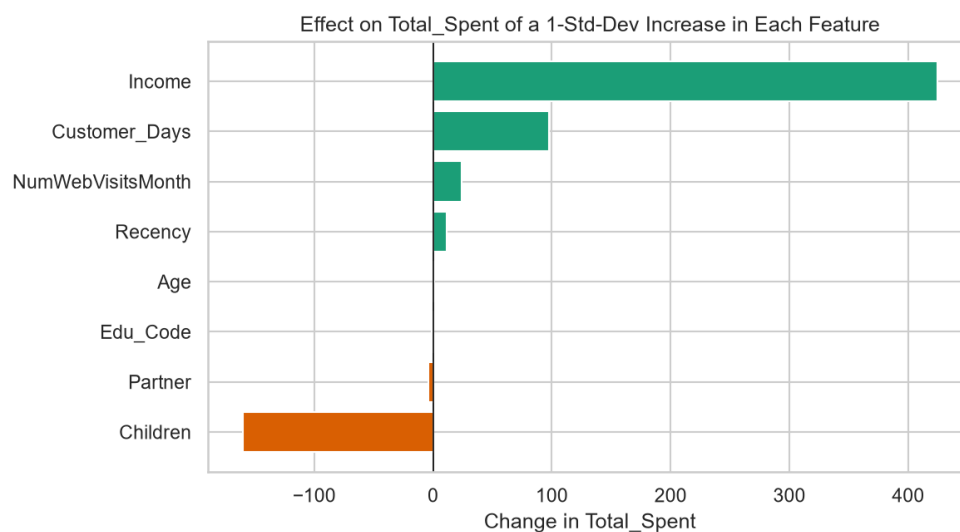


Figure 17: Effect of a one-standard-deviation increase in each feature

Income is the strongest driver (+1,000 income $\approx$ +20 spending) and each additional child reduces spending by about 217. Each year as a customer adds about 175. The multiple model

explains **71%** of the variation in spending, and the cross-validated R^2 (0.70) shows the model is stable.



Figure 18: Regression diagnostics: actual vs predicted, residuals

Limitations: the residual plot shows a curved pattern and increasing spread for larger predictions (heteroscedasticity), and some predictions for low-income customers are negative (clipped to zero in the dashboard). Modelling $\log(\text{spending})$ or a non-linear model could improve this.

9. Customer Segmentation using K-Means

Seven features – Income, Total_Spent, Age, Children, Total_Purchases, Recency and NumWebVisitsMonth – were standardised and clustered with K-Means. Standardisation is essential because K-Means uses Euclidean distance.

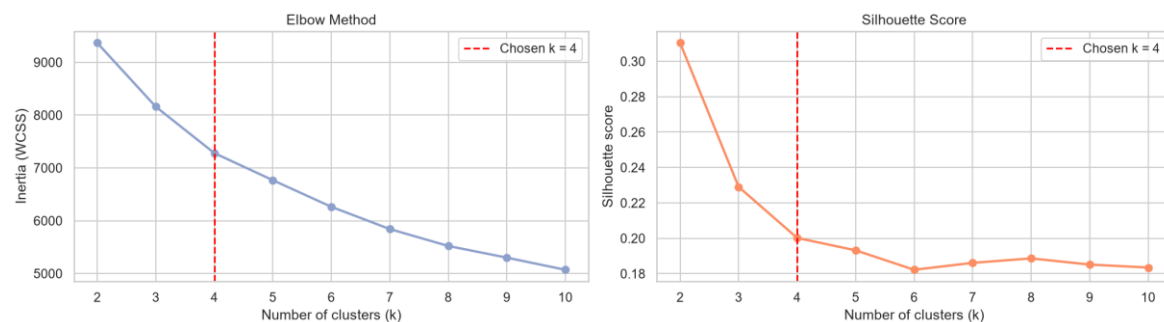


Figure 19: Elbow method and silhouette score

The elbow curve bends at $k = 4$ (the drop in inertia falls from about 880 for $k = 3 \rightarrow 4$ to about 510 for $k = 4 \rightarrow 5$). Although $k = 2$ has the highest silhouette score (0.31), it only separates high and low spenders, which is too coarse for marketing. From $k = 4$ the silhouette is almost flat (≈ 0.20), so **$k = 4$** was selected.

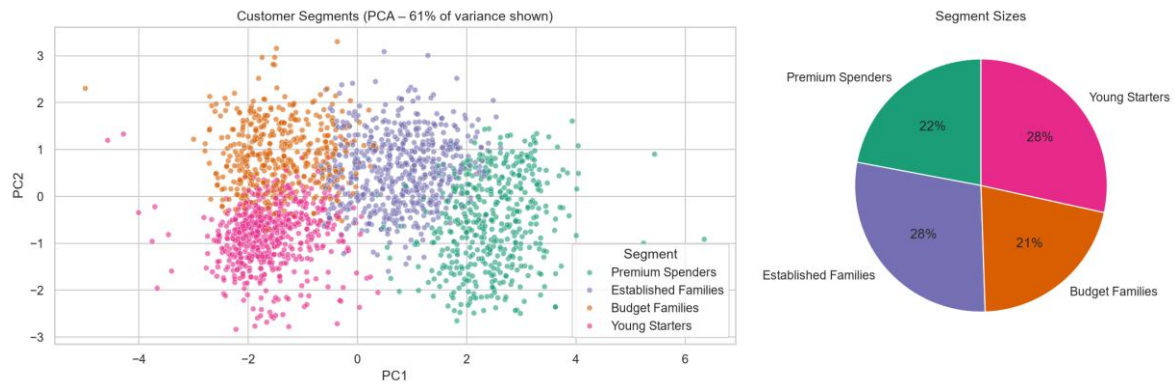


Figure 20: Customer segments in two dimensions (PCA, 61% of variance)

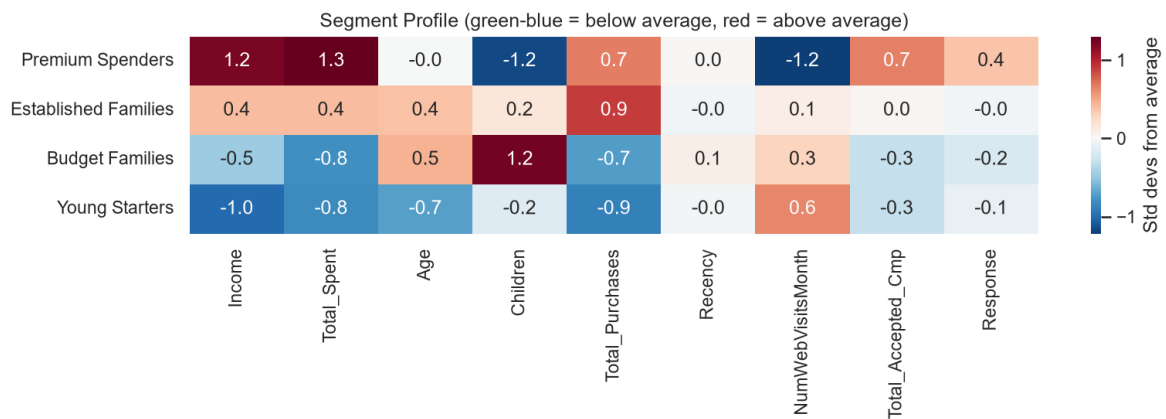


Figure 21: Segment profiles relative to the overall average

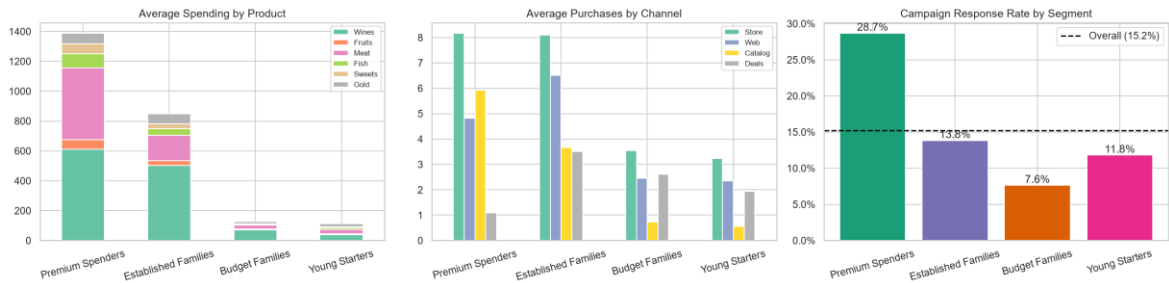


Figure 22: Spending, channels and response rate by segment

Table 10: Customer segments

Segment	Customers	Profile	Revenue share	Response rate
Premium Spenders	453 (22%)	Income $\approx$ 78k, almost no children, catalog buyers, rarely use deals	50.5%	28.7%
Established Families	585 (28%)	Income $\approx$ 61k, age $\approx$ 50, teenagers at home, longest tenure, heavy deal users	39.8%	13.8%
Budget Families	432 (21%)	Income $\approx$ 42k, $\approx$ 2 children, few purchases	4.4%	7.6%
Young Starters	584 (28%)	Age $\approx$ 37, income $\approx$ 30k, visit the website most but buy little	5.3%	11.8%

Table 11: Segment strategies

Segment	Recommended marketing strategy
Premium Spenders	Luxury wine and meat offers, VIP/loyalty programme, catalog campaigns;

	include in every campaign.
Established Families	Value bundles and deals, store and catalog promotions, upgrade offers towards premium products.
Budget Families	Discount coupons, family and kids' bundles, in-store promotions; keep contact cost low.
Young Starters	Web and mobile promotions, first-purchase coupons and retargeting to convert browsing into buying.

Key finding: Premium Spenders and Established Families make up 50% of customers but generate about **90% of revenue** and **67% of all campaign responders**.

10. Classification – Predicting Campaign Response

10.1 Approach

- **Features:** 22 customer attributes (demographics, spending, channels, recency, tenure, previous acceptances, complaints) plus the segment from K-Means (one-hot encoded) – 26 features in total.
- **Stratified 80/20 split:** 1,643 training and 411 test customers, both with 15.2% responders.
- **Class imbalance:** `class_weight = "balanced"`, so errors on the rare responder class cost more.
- **Tuning:** GridSearchCV with 5-fold stratified cross-validation on the training set; the best model is chosen by cross-validated ROC-AUC, keeping the test set for the final unbiased evaluation.
- **Baseline:** predicting "No" for everyone gives 84.7% accuracy but 0% recall, so recall, precision, F1 and ROC-AUC are used instead of accuracy alone.

10.2 Results

Table 12: Classification model comparison (test set, 411 customers)

Model	CV AUC	Accuracy	Precision	Recall	F1	Test AUC
Logistic Regression	0.892	0.791	0.408	0.810	0.543	0.885
Decision Tree	0.825	0.723	0.329	0.778	0.462	0.788
Random Forest	0.885	0.837	0.478	0.683	0.562	0.875

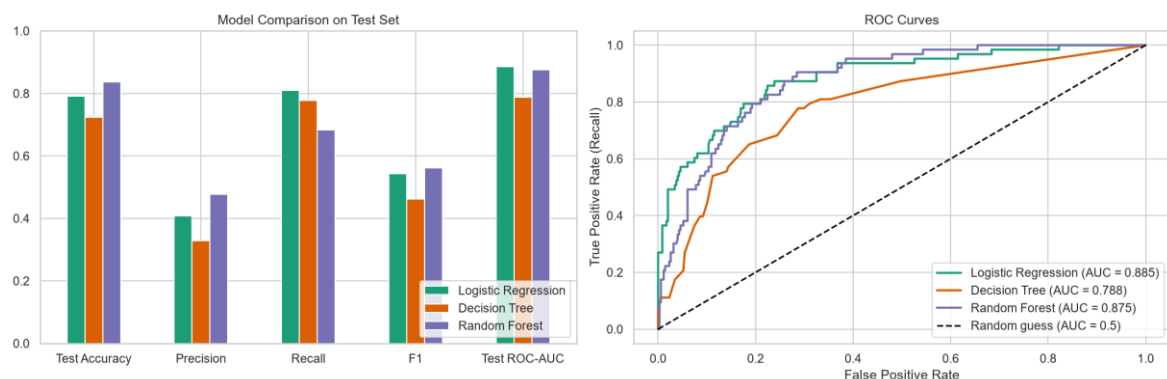


Figure 23: Model comparison and ROC curves

Logistic Regression is the best model (CV AUC 0.892, test AUC 0.885) and has the highest recall (81%). Random Forest is close (AUC 0.875) with higher precision but lower recall. The Decision Tree is weakest (AUC 0.788) but easiest to explain. The simpler linear model performs best because the main relationships are close to linear and additive, and the dataset ($\approx 2,000$ customers) offers limited extra structure for a complex ensemble to exploit.

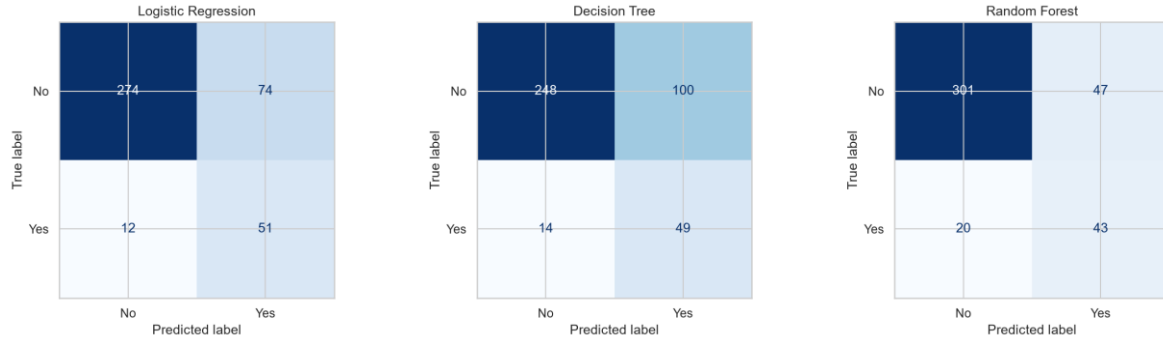


Figure 24: Confusion matrices

The Logistic Regression model correctly finds 51 of the 63 responders in the test set (12 missed) and correctly skips 274 of 348 non-responders, with 74 false positives. For marketing, a missed responder (lost sale) usually costs more than one extra contact, so high recall with moderate precision is the preferred trade-off.

10.3 Drivers of Campaign Response

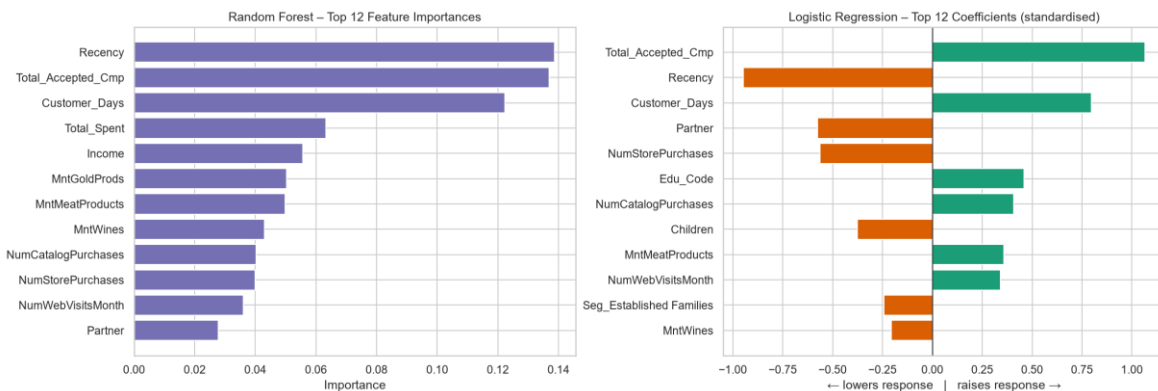


Figure 25: Random Forest feature importance and Logistic Regression coefficients

- **Previous campaign acceptances (+):** odds of responding $\times 2.9$ per standard deviation – past responders respond again.
- **Recency (-):** recent buyers are much more likely to respond (odds $\times 0.39$ per standard deviation of more days).
- **Customer tenure (+):** long-standing customers respond more (odds $\times 2.2$).
- **Partner and children (-):** singles and customers without children respond more.
- **Catalog purchases, meat spending and education (+)** increase response.

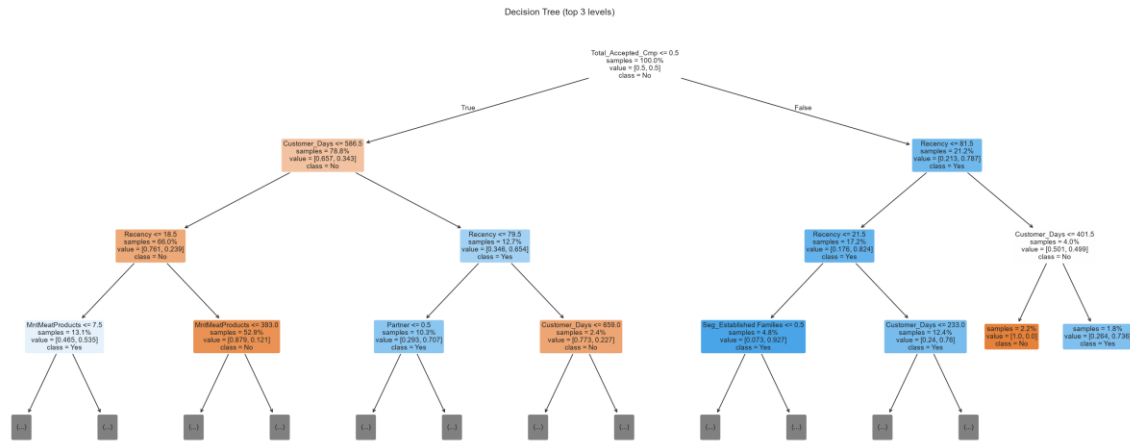


Figure 26: Decision tree (top three levels)

10.4 Business Impact – Targeting Simulation

Table 13: Cumulative gains on the test set (Logistic Regression)

Contact top	Responders captured	Response rate of contacted	Lift
10%	49.2%	75.6%	4.9×
20%	68.3%	52.4%	3.4×
30%	79.4%	40.7%	2.7×
40%	87.3%	33.5%	2.2×
50%	93.7%	28.6%	1.9×
100%	100.0%	15.3%	1.0×

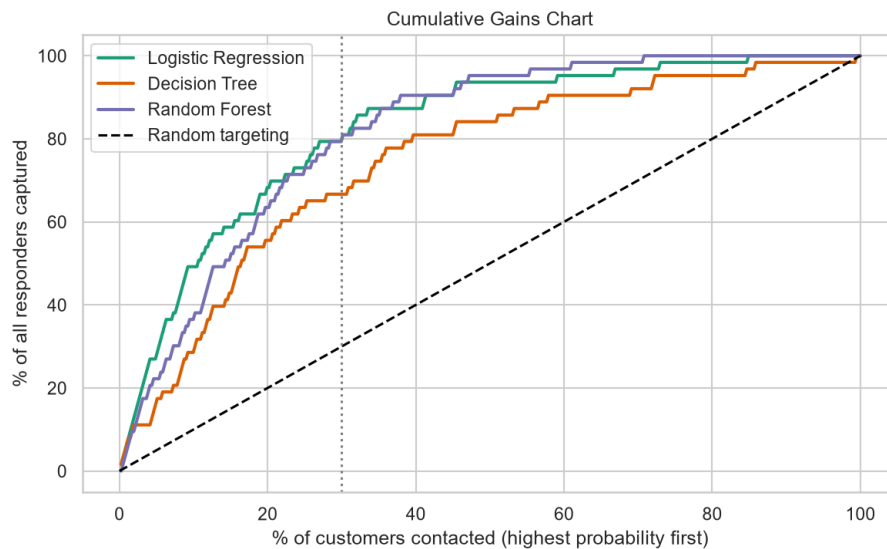


Figure 27: Cumulative gains chart

Contacting only the **top 30%** of customers ranked by the model captures about **80% of responders** at a response rate of about 41% (2.7× better than random), while reducing the number of contacts by 70%. Using the dataset's own cost (3 per contact) and revenue (11 per response), contacting all 2,054 customers produces a loss of 2,719, whereas contacting the model's top 30% produces a profit of 979.

11. Application Development – Interactive Dashboard

An interactive dashboard was developed in **Streamlit** with **Plotly** charts so that marketing managers can use the analysis without writing code. It loads the cleaned data and trained models and has five pages, with a sidebar filter to focus on selected segments.

Table 14: Dashboard pages

Page	Purpose
Overview	KPI cards (customers, income, spending, revenue, response rate), revenue by product, channels, campaign acceptance
Customer Segments	Segment cards, PCA plot, response by segment, product and channel mix, segment explorer with strategy
Model Performance	Model comparison table, ROC curves, confusion matrices, feature importance, regression results
Predict a Customer	Form to enter a customer's details; returns segment, response probability, expected spending and a campaign decision
Campaign Targeting	Choose the % of customers to contact, cost and revenue; shows responders captured, profit, gains chart and a downloadable target list

Customer Intelligence

Navigate

Overview

Customer Segments

Model Performance

Predict a Customer

Campaign Targeting

Filter segments

Premium Spenders

Established Families

Budget Families

Young Starters

Dataset: Customer Personality Analysis (Kaggle) - 2,054 customers after cleaning

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Intelligent Customer Segmentation & Campaign Optimization

Understand customers, find the right segments and target the people most likely to respond.

Customers
2,054
100% of total

Average income
52,036
per year

Average spending
606
last 2 years

Total revenue
1.25 M
last 2 years

Response rate
15.2%
last campaign

Revenue by Product Category

Product Category	Percentage
Wines	50.1%
Meat	27.7%
Gold	7.23%
Fish	6.16%
Sweets	4.48%
Fruits	4.32%

Purchases by Channel

Channel	Number of purchases
Store	~11,500
Web	~8,500
Catalog	~5,500
Deals	~4,500

Income vs Total Spending

Legend: Premium Spenders (green), Established Families (blue), Budget Families (orange), Young Starters (pink).

Acceptance Rate of Each Campaign

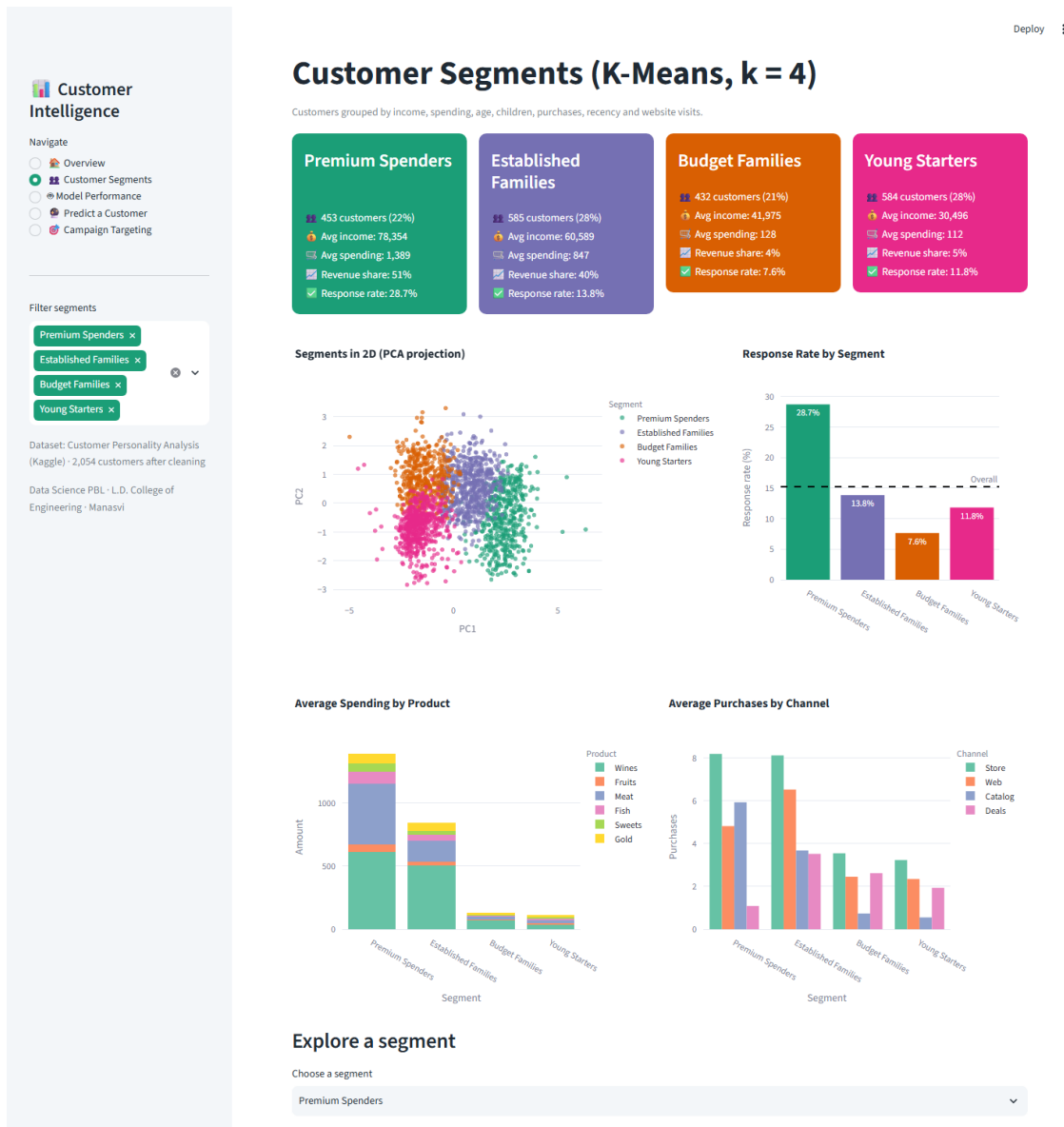
Campaign	% accepted
Cmp 1	6.6%
Cmp 2	1.3%
Cmp 3	7.4%
Cmp 4	7.7%
Cmp 5	7.2%
Last	15.2%

Key insights

- Wine and meat generate about 78% of revenue.
- Customers without children spend ~6x more than parents, and income is strongly linked to spending ($r = 0.78$).
- Only 15% of customers responded to the last campaign - 85% of contacts were wasted.
- Past responders, recent buyers and long-standing customers are the most likely to respond again.

Figure 28: Dashboard – Overview page

21



Customer Intelligence

Navigate

Overview

Customer Segments

Model Performance

Predict a Customer

Campaign Targeting

Filter segments

Premium Spenders

Established Families

Budget Families

Young Starters

Dataset: Customer Personality Analysis (Kaggle) - 2,054 customers after cleaning

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Model Performance

Response classification Spending regression

Best model: Logistic Regression (chosen by 5-fold cross-validated ROC-AUC on the training set). Evaluated on a stratified 20% test set (411 customers).

Model	CV ROC-AUC	Test Accuracy	Precision	Recall	F1	Test ROC-AUC
Logistic Regression	0.892	0.791	0.408	0.810	0.543	0.885
Decision Tree	0.825	0.723	0.329	0.778	0.462	0.788
Random Forest	0.885	0.837	0.478	0.683	0.562	0.875

ROC Curves

Confusion matrix for

Logistic Regression

Confusion Matrix - Logistic Regression

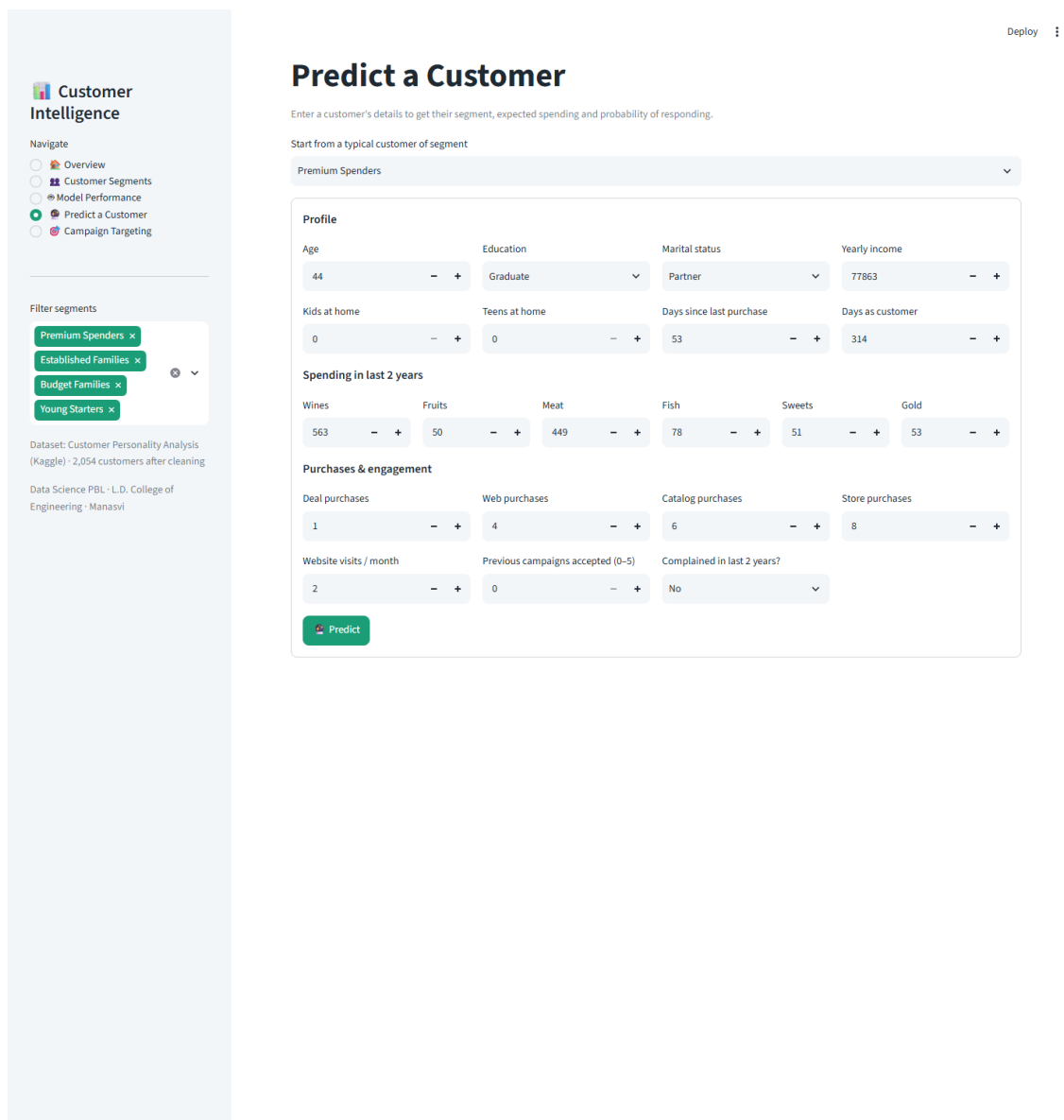
Actual No	274	74
Actual Yes	12	51
	Predicted No	Predicted Yes

Random Forest - Feature Importance

Logistic Regression - Coefficients

Figure 30: Dashboard – Model Performance page

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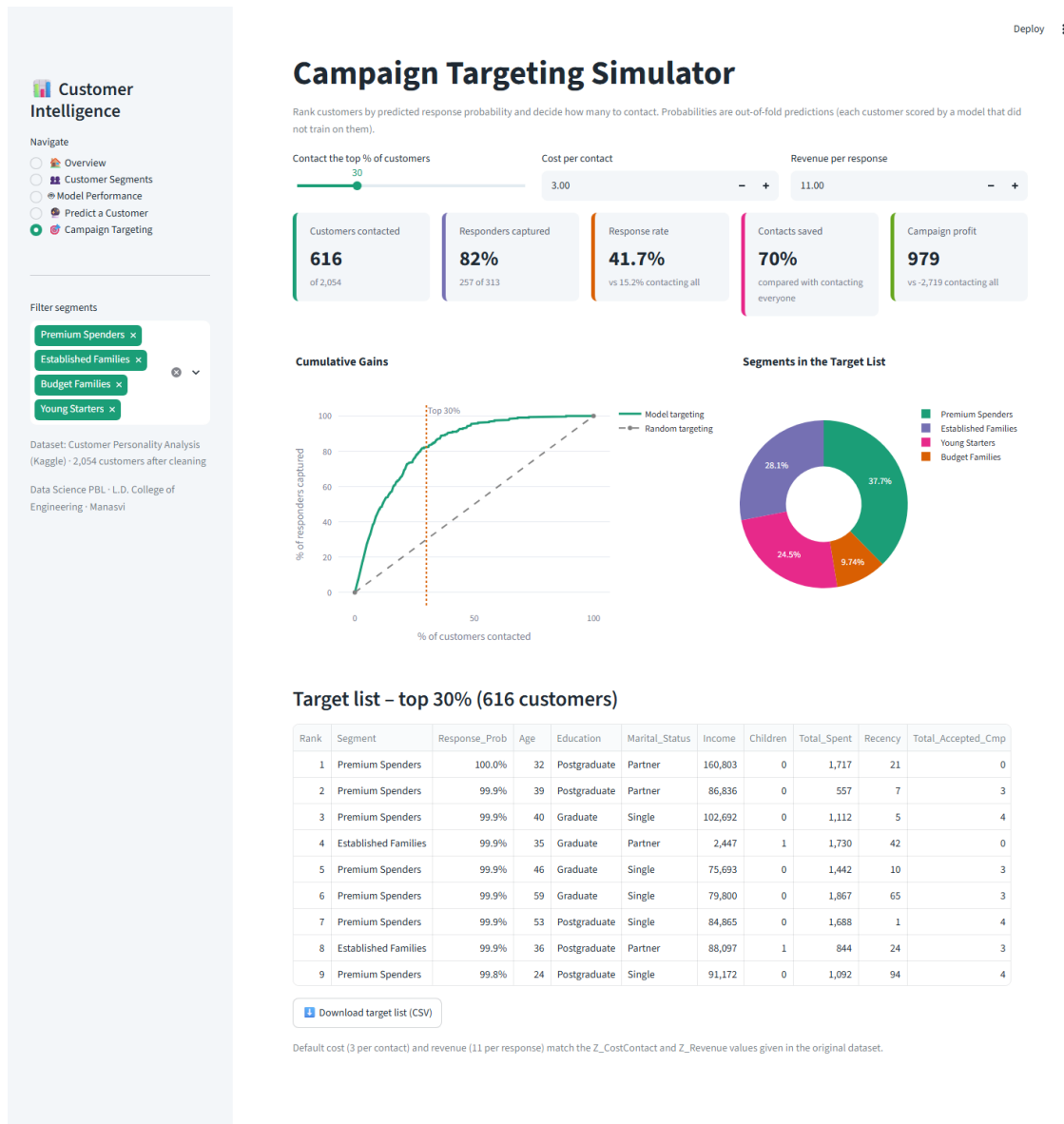


Figure 32: Dashboard – Campaign Targeting page

12. Business Insights and Recommendations

- **Stop mass campaigns.** With a 15% response rate, contacting everyone loses money. Rank customers with the response model and contact only the top 30–40%.
- **Prioritise Premium Spenders and Established Families.** They generate about 90% of revenue and two-thirds of responders; offer premium wine and meat products, loyalty rewards and value bundles.
- **Re-target past responders and recent buyers first.** Previous acceptance and recency are the strongest predictors of response.
- **Use low-cost digital channels for Budget Families and Young Starters,** such as e-mail coupons and web promotions, since their response and spending are low.
- **Focus product campaigns on wine and meat,** which make up 78% of revenue, and use smaller categories for cross-selling bundles.
- **Run pilot campaigns of about 550 customers** to estimate the response to a new offer within $\pm 3\%$ before a full launch.

13. Conclusion and Future Scope

13.1 Conclusion

This project built a complete customer intelligence system on a real retail dataset. Data cleaning and EDA revealed that spending is driven by income and family situation and that only 15% of customers respond to campaigns. Sampling and estimation quantified key business measures with confidence intervals and confirmed the main differences between responders and non-responders statistically. Linear regression explained 71% of the variation in customer spending. K-Means clustering identified four actionable segments, and Logistic Regression predicted campaign response with a ROC-AUC of 0.885. Model-based targeting can capture about 80% of responders with 70% fewer contacts, turning an unprofitable campaign into a profitable one. The Streamlit dashboard makes these insights directly usable by business stakeholders.

13.2 Limitations

- The data comes from one company and one time period, so results may not generalise to other businesses.
- Income is self-reported and some customers have unusual combinations (e.g., very high income with almost no spending).
- Linear regression assumptions (linearity, constant variance) are only partly satisfied.

13.3 Future Scope

- Try gradient boosting models (e.g., XGBoost) and resampling methods such as SMOTE.
- Calibrate predicted probabilities and choose the decision threshold by expected profit.
- Build product-specific response models (wine, meat, gold) for personalised offers.
- Integrate the dashboard with a live CRM database and deploy it on the cloud for the marketing team.
- Run A/B tests to measure the real uplift of model-based targeting.

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Appendix: Project Structure and How to Run

Source code repository: <https://github.com/manasvigohil28/Custom-Segmentation-Campaign-Optimization>

- `app.py` – Streamlit dashboard
- `utils.py` – shared cleaning and feature engineering
- `notebooks/analysis.ipynb` – complete analysis (data cleaning to classification)
- `data/` – original and processed datasets
- `models/` – trained models (.pkl) and metrics.json
- `figures/` – all charts
- `report/` – project report and dashboard screenshots
- `requirements.txt` – Python packages

Run: `pip install -r requirements.txt`, then `streamlit run app.py` (opens at <http://localhost:8501>).